INFERENCE TIME ADAPTATION FOR IMPROVED RETINAL DISEASE DIAGNOSIS USING OPTICAL COHERENCE TOMOGRAPHY IMAGES

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ABSTRACT

Spectral-Domain Optical Coherence Tomography (SDOCT) is the most common and cost-effective variant of OCT for diagnosing retinal diseases. Deep learning models that are utilized for automated diagnosis of these images often exhibit poor performance - across data-sets with varying acquisition settings, limiting their clinical utility. This work, for the first time proposes "Inference Time Adaptation (ITA)" with lightweight deep learning models for SDOCT analysis, enhancing model deployability and generalizability. Furthermore, this work extends ITA, introducing "Inference Time Adaptation with Domain Transforms (ITADT)" to improve the generalizability of deep learning models for retinal disease diagnosis. Domain specific augmentations are applied to test samples to enable regularized model adaptation, enhancing classification performance. The proposed approach is model agnostic and can be applied to any trained deep learning model. The proposed methods achieve significant performance improvement of $>10\%$ on average in accuracy, compared with the un-adapted/original deep learning model on benchmark data-sets for classification of retinal diseases.

Index Terms— Optical Coherence Tomography, Deep Learning, Generalizability, Inference Time Adaptation, Retinal Diseases

1. INTRODUCTION

Optical coherence tomography (OCT) is the most widely used imaging modality for diagnosing retinal diseases [1]. The current OCT systems can be broadly categorized into two types: time-domain OCT (TD-OCT) and frequency-domain OCT (FD-OCT) [2]. TD-OCT uses a low-coherence light source and measures the time delays in the echoes of the reflected signal to determine the intensities at various tissue depths. FD-OCT relies on measuring the interference spectrum using a high-speed spectrometer to resolve the tissue depth information. FD-OCT is further categorized as spectral-domain OCT (SD-OCT) and swept-source OCT (SS-OCT) [2]. SD-OCT imaging systems are often preferred because of their speed and cost-effectiveness. There is an increased use of deep learning models for retinal disease diagnosis owing to their performance [3–5]; however, these models still lack deployability and generalizability across multiple sites. The lack of generalizability can be attributed to the induced speckle noise [6] owing to variations in imaging protocols/scanners. Methods such as attention based networks [7], ensemble models [5], adversarial adapted models [8], and noise regularized models [9] improve the generalizability of these models for retinal disease diagnosis. However, these methods build regularized deep models from training data and do not generalize specifically to the test sample of an OCT B-scan image.

Recently, inference (test) time adaptation of models has attracted considerable attention in computer vision for addressing out-of-distribution (OOD) generalization. In this class of approaches, pre-trained model weights are adapted for the target data during the inference time. In Ref. [10], authors propose test-time training with self-supervised computer vision tasks, which are also learnt during the training phase. One drawback of this approach is the lack of suitability for off-the-shelf adaptation because the training procedure must be augmented with self-supervision tasks. A full test-time adaptation (TENT) has been proposed in Ref. [11] for computer vision tasks, where only pre-trained model and target data was required for adaptation, with no change in training procedure. Weight modification via entropy minimization has shown to correspond to confidence maximization, yielding a state-of-the-art performance on standard out-of-distribution datasets. The adaptations of TENT for the task of medical image segmentation has been demonstrated in Ref. [12].

This study proposes methods for inference/test-time adaptation of lightweight deep learning models for retinal disease diagnosis with the aim of improving model deployability and model generalizability. To the best of our knowledge, this study is the first to address retinal disease classification using a test-time adaptation framework. The key contributions of this work are as follows:

1. This work proposes "Inference Time Adaptation (ITA)" for retinal disease diagnosis by combining the established TENT [11] approach with lightweight deep learning models, enhancing model deployability and improving model

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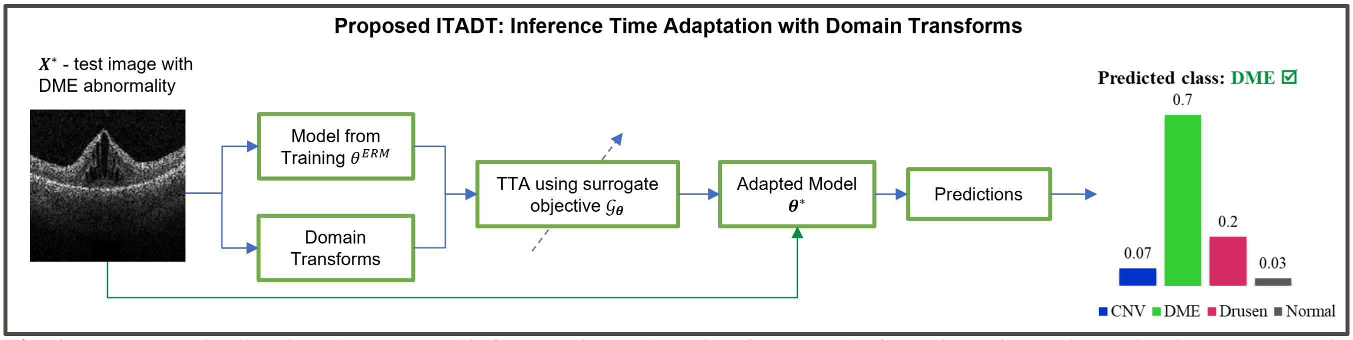

Fig. 1: Key steps highlighting the proposed inference time adaptation frameworks for retinal diagnosis. Switching OFF domain transforms results in ITA. ITADT uses domain transforms to improve on top of ITA.

generalizability.

2. Furthermore, this work extends ITA, introducing "Inference Time Adaptation with Domain Transforms (ITADT)" to improve the generalizability of deep learning models for retinal disease diagnosis. Domain specific augmentations are applied to test samples to enable regularized model adaptation, enhancing the retinal disease classification performance.
3. Novel loss functions for inference time unsupervised model updation enforcing confidence in predictions while maintaining consistency between original scans and domain transformed scans.
4. Demonstration of significant performance improvement on all metrics of interest, across benchmark data-sets and across different baseline architectures.

2. METHODS

2.1. Preliminaries

In this study, the class of feed-forward neural networks $h_{\boldsymbol{\theta}} : \mathcal{X} \rightarrow \mathcal{Y}$, parameterized by their weights $\boldsymbol{\theta}$, was considered for mapping OCT images to retinal diseases. The training dataset is denoted by $\mathcal{S}$ and consists of m i.i.d. samples: $\{(X_1, Y_1), (X_2, Y_2), \cdots, (X_m, Y_m)\}$. $X_i \in \mathcal{X}$ denotes the i^{th} OCT image and $Y_i \in \mathcal{Y}$ is the corresponding ground-truth class label, which can take values between $\{1, 2, \cdots, C\}$ for C-way classification problem. The parameters $\boldsymbol{\theta}$ are learnt via empirical risk minimization (ERM) process that involves optimizing the empirical risk $J_{\text{ERM}}(\boldsymbol{\theta})$ over the training set $\mathcal{S}$,

$$J_{\text{ERM}}(\boldsymbol{\theta}) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}(h_{\boldsymbol{\theta}}(X_i), Y_i), \tag{1}$$

where $\mathcal{L}$ is a suitable sample-wise loss function, usually cross-entropy or variants, that measures the distance between predicted output $h_{\boldsymbol{\theta}}(X_i)$ and Y_i. The neural network weights obtained by optimizing Eq. (1) are denoted by $\boldsymbol{\theta}_{\text{ERM}}$, which is the output of a typical training procedure.

2.2. Proposed Method

2.2.1. *ITA: Inference Time Adaptation for Retinal Disease Diagnosis*

ITA combines the established TENT [11] approach with lightweight deep learning models (ResNet18 [13] and MobileNetV2 [14]) for retinal disease diagnosis. Let f be a lightweight deep learning model with a set of parameters $\boldsymbol{\theta}$ (pre-trained for retinal disease prediction) and $f_{\boldsymbol{\theta}}(X^*)$ be the vector that denotes softmax probabilities corresponding to C classes. For ease of notation, the likelihood of class k obtained by inferring X^* using $\boldsymbol{\theta}$ is denoted by $y_i^k = f_{\boldsymbol{\theta}}(X^*)[k]$. The uncertainty in the predictions can be calculated using a suitable function such as entropy. As shown in Eq. (2), by minimizing the entropy, confidence of the predictions can be maximized.

$$\mathcal{L}_{ITA} = -\sum_{k=1}^{C} y_i^k \cdot \log y_i^k. \tag{2}$$

2.2.2. *ITADT: Inference Time Adaptation with Domain Transforms*

Clinical experts often make diagnostic decisions by utilizing an additional source of information, or by applying a suitable transformation to medical images for the current diagnostic task. Common examples of such transformations include changing the window level in CT, using all available parametric images in MRI, post-processing knobs such as gain, and TGC in ultrasound. These transformations are *modality- and task-specific*. In this work, in contrast to the common augmentations found in the deep learning literature, such as geometric or intensity transforms, modality-specific transforms are referred to as *domain-transforms*. The goal of the proposed work is to improve the generalization of deep learning models on OCT images across varying SNR levels, the chosen domain transforms are simple denoisers. Common OCT denoisers are used as domain transforms, including block matching and 3D filtering (BM3D), non-local means filtering (NLM), and bilateral filtering (Bilat). The utilization of a group of denoisers provides complementary denoising characteristics and overcomes the relative disadvantages of the individual filters. For rest of this work, the input image and domain transforms are denoted by $\{T_i(X^*)\}_{i=0,1,2,3}$, where $T_0(X^*)$ is the identity mapping (original image) and the remaining $T_{i=1,2,3}$ corresponds to the denoiser output.

A novel two-part composite loss function utilizing the original OCT images, domain transforms, and a pre-trained model was proposed to act as the surrogate $\mathcal{G}_{\boldsymbol{\theta}}$ to perform unsupervised model adaption during the inference time. These two parts adapt the models such that the confidence of the predictions for each of the domain transforms is improved, while their predictions agree with each other. Intuitively, the adaptation procedure attempts to mimic expert behavior, where an accurate diagnosis arrives only when the confidence and consistency of the diagnosis on available information is maintained. The mathematical framework for these loss functions is as follows:

Modeling Confidence Loss ($\mathcal{L}_{conf}$): Let $f_{\boldsymbol{\theta}}(T_i(X^*))$ be the vector that denotes softmax probabilities corresponding to C classes. For ease of notation, the likelihood of class k obtained by inferring $T_i(X^*)$ using $\boldsymbol{\theta}$ is denoted by $y_i^k = f_{\boldsymbol{\theta}}(T_i(X^*))[k]$. The uncertainty in the predictions can be calculated using a suitable function such as entropy. As shown in Eq. (3), by minimizing the entropy, confidence of the predictions can be maximized. Since the aim is to increase confidence on original image as well as domain transformed ones, $\mathcal{L}_{conf}$ is defined as shown in Eq. (4).

$$\mathcal{L}_{ent}(T_i(X^*), \boldsymbol{\theta}) = -\sum_{k=1}^{C} y_i^k \cdot \log y_i^k \tag{3}$$

$$\mathcal{L}_{conf} = \frac{1}{N+1} \sum_{i=0}^{N} \mathcal{L}_{ent}(T_i(X^*), \boldsymbol{\theta}) \tag{4}$$

Modeling Consistency Loss ($\mathcal{L}_{cons}$): As noted earlier, the second part of loss function aims to minimize disagreements between predictions on different members of domain transforms. With categorical cross-entropy as the proposed choice of loss function, measure of disagreement between predictions on T_i and T_j can be modeled as,

$$\mathcal{L}_{CE}(T_i(X^*), T_j(X^*), \boldsymbol{\theta}) = -\sum_{k=1}^{C} y_i^k \cdot \log y_j^k. \tag{5}$$

To ensure that consistency is maintained across all domain transforms an aggregate of pairwise disagreements was proposed and can be defined as,

$$\mathcal{L}_{cons} = \frac{1}{N(N+1)} \sum_{i=0}^{N} \sum_{j=0, i\neq j}^{N} L_{CE}(T_i(X^*), T_j(X^*), \boldsymbol{\theta}). \tag{6}$$

Finally utilizing Eq. (4) and (6), the label-free unsupervised, regularized surrogate loss function for inference time adaptation can be defined as,

$$\mathcal{L}_{ITADT} = \mathcal{L}_{conf} + \lambda \cdot \mathcal{L}_{cons}, \tag{7}$$

where λ is the regularization parameter.

3. EXPERIMENTS

3.1. Datasets

The details of the datasets utilized in this study are as follows:

1. NEH Dataset [17]: has a total of 148 subject studies with approximately 20-60 B-mode scans obtained per subject. The images were acquired from Heidelberg Spectralis imaging system with axial resolution of 3.5μm, with non-uniform lateral and azimuth resolution across different subjects.
2. DHU Dataset [18]: constitutes 45 subject volumes obtained from Heidelberg Spectralis system with axial resolution of 3.85μm and lateral resolution ranging from 6-12μm. The number of B scans per participant ranged from to 31-97. Both NEH and DHU datasets contain images from 3 classes: AMD, DME and Normal.

3.2. Implementation Details

Experiments were conducted using the PyTorch framework on a Linux workstation with 128 GB of CPU RAM and two NVIDIA Quadro RTX 8000 GPU cards with 48 GB each. The pre-trained models were developed in close alignment with the approach detailed in [9]. The test-time adaptation procedures of ITA and ITADT approach were run with a batch size of 256 images, number of steps $t_{max} = 4$ and learning rate $\alpha = 1 \times 10^{-2}$ using the Adam optimizer. The regularization parameter λ in Eq. 7 was set to 1. Similar to TENT, ITA adapts only batchnorm parameters. However, ITADT adapts all the trainable parameters. To analyze the classification performance, standard metrics, including the F1-score, accuracy, and recall, were compared.

4. RESULTS AND DISCUSSION

Proposed ITA and ITADT outperform state-of-the-art for retinal disease classification. Table 1 reports improved performance of ITA and ITADT over state-of-the-art methods. ITADT improves over different types of architectures, including larger models [4, 15] and lightweight CNN [16]. It is pointed out that ITADT outperforms the recently proposed noise-regularized training as well [9] by a significant margin of 11% in the F1-score. Further, architecture agnostic improvements with respect to baseline were shown in Table 2. The proposed domain transforms for retinal disease diagnosis

Table 1: Comparisons with state-of-the-art retinal disease classification methods. Rows 1-3: Models trained on $\sim 60\%$ of OCT B-scans from the NEH dataset and tested on all ~ 3000 images of the DHU dataset. Rows 4-6: Models trained on $\sim 60\%$ of OCT B-scans from the DHU dataset were tested on all ~ 4000 images of the NEH dataset. The best results are shown in bold. ITA and ITADT were built on ResNet18.

Study	Metric $\mid$ Method	Feng et al. [15]	Lu et al. [4]	Sunija et al. [16]	Paluru et al. [9]	ITA (Proposed)	ITADT (Proposed)
NEH $->$ DHU	**Precision**	0.79 ± 0.06	0.74 ± 0.09	0.85 ± 0.02	0.86 ± 0.05	0.87±0.01	**0.88±0.01**
	F1-score	0.76 ± 0.06	0.62 ± 0.17	0.67 ± 0.11	0.84 ± 0.06	0.87±0.01	**0.88±0.01**
	Accuracy	0.76 ± 0.06	0.64 ± 0.17	0.70 ± 0.08	0.84 ± 0.06	**0.88±0.01**	**0.88±0.01**
DHU $->$ NEH	**Precision**	0.57 ± 0.03	0.57 ± 0.06	0.65 ± 0.05	0.66 ± 0.07	0.75±0.02	**0.76±0.02**
	F1-score	0.50 ± 0.02	0.52 ± 0.05	0.60 ± 0.09	0.61 ± 0.06	0.72±0.03	**0.73±0.02**
	Accuracy	0.53 ± 0.03	0.54 ± 0.05	0.61 ± 0.08	0.61 ± 0.06	0.72±0.03	**0.73±0.02**

Table 2: Demonstrating architecture agnostic performance. Rows 1-3: Models trained on $\sim 60\%$ of OCT B-scans from the NEH dataset and tested on all ~ 3000 images of the DHU dataset. Rows 4-6: Models trained on $\sim 60\%$ of OCT B-scans from the DHU dataset were tested on all ~ 4000 images of the NEH dataset. The best results (F1-score) are shown in bold.

Study	Adaptation	ResNet18 [13]	MobileNetV2 [14]
NEH $->$ DHU	No Adaptation	0.70 ± 0.07	0.82 ± 0.03
	ITA	0.87 ± 0.01	$\mathbf{0.84 \pm 0.01}$
	ITADT	$\mathbf{0.88 \pm 0.01}$	$\mathbf{0.84 \pm 0.01}$
DHU $->$ NEH	No Adaptation	0.52 ± 0.01	0.59 ± 0.05
	ITA	0.72 ± 0.03	$\mathbf{0.74 \pm 0.01}$
	ITADT	$\mathbf{0.73 \pm 0.02}$	0.73 ± 0.01

helped the model to alter weights with more representative and discriminative features, as observed in OOD test data, leading to more accurate and confident predictions. The proposed approach also reduces the entropy or variance of the model outputs, which are often used as measures of uncertainty or confidence in the diagnosis. More importantly, this adaptation approach prevents the model from overfitting the training data or forgetting the original knowledge by using regularization to preserve the model's generalization ability and confidence in both in-distribution and OOD data.

The test-time adaptation of deep learning models has shown considerable promise for solving computer vision problems. This popularity can be attributed to TENT [11] which provides a simple and elegant framework for full test-time, off-the-shelf adaptation with no constraints on training procedures. The proposed ITA and ITADT were built on top of TENT, tailoring with lightweight deep learning models and additional regularization by using OCT specific domain transforms. In other words, ITADT proposes a regularized framework for inference–time adaptation using domain transforms. This framework is generic, and can be extended to other problems and modalities. Because the goal of this work was to improve the generalization of retinal diagnosis on OCT B-scans with varying SNR levels, denoisers were chosen as domain transforms.

5. CONCLUSION

Optical Coherence Tomography (OCT) has emerged as a point-of-care imaging modality for diagnosing retinal conditions. Widespread adoption can be attributed, in part, to its cost-effective variant, SD-OCT imaging, which has improved accessibility. Although deep learning algorithms hold promise for SD-OCT based retinal disease classification, their performance drops considerably when applied to noisy scans or datasets from different scanners, as previously reported. This study aimed to enhance the generalizability of deep learning models through inference-time adaptation. This approach also extends its effectiveness to different types of scanners, including Heidelberg Spectralis and Bioptigen, Inc., across diverse acquisition settings. The model-agnostic approach of ITADT and demonstration of the same is a significant development for the generalizability of deep learning models for retinal disease diagnosis.

6. ACKNOWLEDGMENTS

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7. CONFLICTS OF INTEREST

The authors declare that there are no conflicts of interest related to this article.

8. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made publicly available by DHU [17] and NEH [18]. Ethical approval was not required as confirmed by the license attached with the open access data.

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